

## Lab Objectives

- Apply Auto Layout to create responsive UI elements
  - Create and manage Components & Instances
  - Use Component Properties (Boolean, Text, Instance Swap)
  - Build Variant Sets for different UI states
  - Develop a reusable Button System
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### Task 1: Auto Layout

Create a responsive **Tag/Badge** UI element

- **Steps:**
    1. Create a **Text Layer** → Write: Design
    2. Create a **Rectangle (16×16)** → Icon placeholder
    3. Select both → Press **Shift + A** (Apply Auto Layout)
    4. Set properties:
      - Direction: **Horizontal**
      - Gap: **8px**
      - Padding: **12px (horizontal), 6px (vertical)**
    5. Set width to **Hug Contents**
    6. Apply **Corner Radius = 100** (pill shape)
  -  **Checkpoint:**
    - Change text to *"User Experience"*
    - Confirm the container resizes automatically
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### Task 2: Components & Instances

- Create a reusable **Button Component**
  - **Steps:**
    1. Create a button using Auto Layout:
      - Text: Button
      - Padding: **12px / 24px**
    2. Convert to component:
      - Press **Ctrl/Cmd + Alt + K**
    3. Rename:
      - Button/Primary/Default
    4. Open **Assets Panel**
    5. Drag **3 instances** onto canvas
    6. Change text of ONE instance → Save Changes
  -  **Checkpoint:**
    - Only one instance should change
    - Edit master color → ALL instances update
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### Task 3: Component Properties

- Add dynamic controls to your button
  - **Steps:**
  - **1 Boolean Property**
    - Select icon layer
    - Add property → **Boolean** → “Show Icon”
  - **2 Text Property**
    - Select text layer
    - Add property → **Text** → “Label”
  - **3 Instance Swap**
    - Replace icon with icon component
    - Add property → **Instance Swap** → “Icon”
  - **✓ Checkpoint:**
    - Select instance
    - Change **Label from right panel**
    - Toggle icon ON/OFF
    - Master component remains unchanged
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### Task 4: Variants & States

- Create a **Button Variant Set**
  - **Steps:**
    1. Create 4 button styles:
      - Primary Default
      - Primary Hover
      - Secondary Default
      - Secondary Hover
    2. Select all → **Combine as Variants**
    3. Rename properties:
      - Property 1 → **Type**
      - Property 2 → **State**
    4. Assign values:
      - Type: Primary / Secondary
      - State: Default / Hover
    5. Drag instance → test dropdowns
  - **✓ Checkpoint:**
    - Switch between all 4 states using right panel
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**Submit Figma file (Project-Group based: File name==  
Roll#1+Roll#2) on GCR within deadline.**